Personal Statement

Odeyemi Olusegun Israel

UK Global Talent Visa — Digital Technology (Exceptional Promise)

I am an independent researcher and systems engineer based in Derby, United Kingdom. Over the past three years—without institutional affiliation, funding, or supervision—I have built a blockchain-native compliance platform processing 2.64 million transactions, produced three original research papers establishing a unified mathematical theory of instability detection across finance, energy infrastructure, and fluid dynamics, filed a UK patent application covering four inventions, and open-sourced developer tools in TypeScript and Python. I am applying under the Exceptional Promise route because my work demonstrates the capacity to become a leading figure in the UK's applied mathematics, financial technology, and machine learning sectors.

1. What I have built.

My central engineering project is **AMTTP**—the Anti-Money Laundering Transaction Trust Protocol. It addresses a problem that no existing commercial product has solved: enforcing regulatory compliance *before* a blockchain transaction reaches finality, rather than generating alerts after the fact. Chainalysis, Elliptic, and TRM Labs all operate post-settlement; AMTTP intervenes at the protocol level, within the approximately 12-second Ethereum block confirmation window.

The system spans four architectural layers: client SDKs (TypeScript and Python, both open-sourced on GitHub), a compliance orchestration engine combining ML risk scoring, graph-based relationship analysis, sanctions screening, and configurable policy rules, an offline ML training pipeline, and a smart contract layer with 18 contracts deployed on Ethereum Sepolia. The infrastructure comprises 17 containerised microservices coordinated through a central gateway. The ML pipeline uses a Composite Teacher architecture trained on 2.64 million transactions, achieving a production ROC-AUC of **0.9879** and F1 of **0.9012** with 21 features and zero training-serving skew.

I also built **zkNAF**, a zero-knowledge proof framework that allows institutions to verify KYC credentials, risk score ranges, and sanctions non-membership without exposing personal data—resolving the tension between GDPR's data minimisation requirements and AML transparency obligations at the architectural level.

The AMTTP architecture is described in my published paper: *Olusegun Odeyemi, "AMTTP: A Four-Layer Architecture for Deterministic Compliance Enforcement in Institutional DeFi," TechRxiv, February 27, 2026*, DOI: 10.36227/techrxiv.177220113.33816607/v1.

2. What I have discovered.

Alongside the engineering work, I have pursued a research programme that began with a practical question—why do fraud detectors miss what they are designed to find?—and evolved into a unified mathematical theory of instability in complex systems, with results spanning machine learning, systemic financial risk, electrical grid failure, algorithmic stablecoin collapse, and the regularity of the 3D Navier-Stokes equations.

Paper 1: Blind Spot Decomposition Theory (BSDT). Building on the engineering experience of AMTTP (see published paper above), my first research paper provides a mathematical framework for characterising missed fraud. I decompose detection failures into four near-orthogonal components—Camouflage (δ_C), Feature Gap (δ_G), Activity Anomaly (δ_A), and Temporal Novelty (δ_T)—and demonstrate that these account for over 95% of explainable variance in missed-fraud indicators. The correction operators recover recall by up to **84.4 percentage points** without retraining or access to model internals. Validated across 4 model architectures, 7 datasets from 6 domains, and over 36 model–dataset combinations, the best variant achieves mean F1 of 0.787 with 88.1% false positive reduction. Target journal: **IEEE TKDE**.

Paper 2: Universal Deviation Law (UDL). My second paper tackles anomaly detection from first principles. I introduce a multi-spectrum tensor framework with 14 composable operators spanning statistical, geometric, spectral, and information-theoretic families. A 14×14 correlation audit confirms that only 1 pair out of 91 is redundant: each operator captures a genuinely distinct perspective on deviation. The best configuration achieves mean AUC of **0.972** and 91% coverage across five benchmarks—categorically outperforming Isolation Forest (0.788), LOF (0.632), and the current state-of-the-art ECOD (0.921)—with a single hyperparameter set and no per-dataset tuning. I prove five formal theorems providing local distinguishability, global sensitivity, and finite-sample guarantees. Target venue: **NeurIPS 2026**.

Paper 3: Systemic Risk as Geometry (AF+NS merged). My third paper—and the centrepiece of the research programme—establishes a geometric framework linking the BSDT deviation operators to the Lyapunov energy of networked agent dynamics. It proves three theorems (equivalence of the detection functional and the physical potential; a spectral phase transition at a critical manifold; and a closed-form optimal damping policy $\gamma^*(E) = E/(E+\theta)$) and validates them across four domains:

- **Banking** (140 quarters of FDIC call-report data): 6-quarter lead before the 2008 GFC using a threshold trained exclusively on pre-2006 data; AUROC = 0.805 at institution level; a 24-institution global panel spanning US, UK, EU, Asia, and Africa achieves AUROC = 0.811 with a 19-quarter GFC lead.
- **Algorithmic stablecoins** (Terra/Luna UST, May 2022): detection at hour 23, one hour after the initial depeg event, with simulation-to-reality correlation $r =$

0.996. Adaptive friction limits the depeg to 0.28% (versus >99% collapse without intervention).

- **Energy infrastructure** (Texas ERCOT, February 2021): detection at hour 24, a full +72 hours before rolling blackouts begin. Adaptive friction reduces peak capacity loss by 71%. The per-agent decomposition reveals that the cascade origin shifted from frozen wind turbines (the public narrative) to gas supply chain disruption (the structural vulnerability)—a finding invisible to event-study analysis.
- **Fluid dynamics** (3D incompressible Navier-Stokes): reinterpreting γ^* as a state-dependent viscosity yields the **P/D halving principle**—adaptive viscosity halves the enstrophy production-to-dissipation ratio at every Reynolds number, confirmed numerically across $\mathrm{Re} \in [314, 6283]$. This provides a new conditional regularity criterion connected to the Clay Millennium Prize Problem and identifies three strategies toward closing the gap to unconditional regularity.

In all three non-financial domains, the dominant BSDT channel is δ_C (camouflage)—the mechanism by which an unstable system conceals its proximity to collapse—demonstrating that the mathematical structure is genuinely universal, not fitted to any single domain. Target journal: **SIAM Journal on Applied Mathematics**.

3. Intellectual property.

On 4 March 2026, I filed a UK patent application with the Intellectual Property Office (Patents Act 1977) covering four inventions unified by the blind-spot energy functional: AMTTP's pre-settlement compliance architecture, the Adaptive Friction stabilisation mechanism, the Universal Deviation Law's multi-spectrum tensor framework, and the BSDT failure-mode correction operators. The specification describes the combined system as a *"State-Adaptive Compliance Enforcement System with Multi-Spectrum Anomaly Detection and Failure-Mode Correction for Decentralised Finance Networks."*

4. Open-source contributions and tools.

The full codebase—research code, pipeline implementations, client SDKs, smart contracts, and replication packages for all three papers—is publicly available on GitHub at `github.com/segetii/fintech`. Specific open-source components include:

- TypeScript and Python client SDKs for AMTTP integration;
- A MetaMask Snap for browser-based compliance verification;
- A GravityEngine simulation toolkit for networked agent dynamics;
- A 3D pseudo-spectral Navier-Stokes solver with BSDT diagnostic suite and adaptive viscosity;
- Complete replication packages for all empirical results.

5. Why the UK, and why now.

The UK is positioning itself as the global leader in responsible fintech regulation. The FCA's evolving framework for crypto-assets will create substantial demand for compliance infrastructure that operates at the protocol level. AMTTP is, as far as I am aware from surveying the commercial landscape, the only system that provides pre-settlement AML enforcement on blockchain transactions.

The cross-domain reach of my research—from banking regulation to energy grid resilience to pure mathematics—aligns with the UK's strategic priorities in financial stability (Bank of England, PRA), infrastructure resilience (Ofgem, National Grid ESO), and mathematical sciences (EPSRC, the Heilbronn Institute). The Navier-Stokes extension, in particular, connects directly to the UK's world-leading position in applied mathematics and fluid dynamics research.

My immediate plans are: (1) submit the three papers to their target journals (IEEE TKDE, NeurIPS 2026, SIAM J. Applied Math); (2) prepare a short letter for *Nature Communications* summarising the cross-domain universality result for broader scientific visibility; (3) commercialise AMTTP as a compliance-as-a-service product for UK-regulated financial institutions and virtual asset service providers; and (4) build a research-led company at the intersection of machine learning, financial regulation, and mathematical stability theory—headquartered in the UK.

I have done this work alone, without a university laboratory, a GPU cluster, or a research budget. In three years, I have gone from a practical question about missed fraud to a unified mathematical framework that detects collapse across banking systems, stablecoins, power grids, and turbulent fluids—and I have built the production engineering to deploy it. What I bring is the ability to move between rigorous mathematical theory and production-grade systems, and the discipline to validate claims before making them. I believe I can contribute meaningfully to the UK's digital economy, scientific reputation, and regulatory leadership, and I am asking for the opportunity to do so.